Project: Worked examples
Title: Glaser condensation (interstitial and surface)
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1. Scope and method

Calculates interstitial condensation in a layered wall by the Glaser method over twelve monthly climate steps. It gives the condensate on each interface, the annual balance, the peak against the limit and the surface temperature factor.

WHAT IT CALCULATES	RELATION
Layer resistance and diffusion thickness	$R_j = d_j/\lambda_j$; $s_{d,j} = \mu_j \cdot d_j$
Total resistance, U-value and interface temperature	$R_T = R_{si} + \Sigma\, R_j + R_{se}$; $U = 1/R_T$; $T_k = T_{int} - (T_{int} - T_e)\cdot(R_{si} + \Sigma_{(j\le k)}\, R_j)/R_T$
Saturation pressure (Magnus)	$p_{sat} = 610.5\cdot\exp(17.269\cdot T/(237.3 + T))$ over water ($T \ge 0$ °C); $p_{sat} = 610.5\cdot\exp(21.875\cdot T/(265.5 + T))$ over ice ($T < 0$ °C)
Vapour pressures	$p_e = RH_e\cdot p_{sat}(T_e)$; $p_i = RH_i\cdot p_{sat}(T_{int})$, or $p_e + f_{\Delta p}\cdot\Delta v(T_e)\cdot R_v\cdot(T_i + T_e)/2$ in the moisture-excess mode with R_v = 461.5 J/(kg·K)
Vapour flux and condensate	$g = \delta 0\cdot\Delta p/\Delta s_d$ with $\delta 0 = 2\times10^{-10}$ kg/(m·s·Pa); $g_c = g_{in} - g_{out}$ on the tangent-construction profile; Mc accumulates $g_c\cdot t$ over 12 months, a wet interface drying at $g_c < 0$
Refinement check	the same year with every layer split into 20 equal sub-layers of the same λ and μ; the annual balance passes only when it passes both ways
Annual balance and peak	PASS when no condensate remains after 12 months; Mc_{max} = the most held on any interface, checked against Mc_{limit}
Surface factor	$f_{Rsi} = 1 - R_{si,f}/R_{T,f}$; $f_{Rsi,min}$ = the largest monthly $(T_{si,min} - T_e)/(T_{int} - T_e)$ with $p_{sat}(T_{si,min}) = p_i/\varphi_{crit}$; PASS when $f_{Rsi} \ge f_{Rsi,min}$

Basis: Hygrothermal performance of building components and elements — internal surface temperature and surface condensation[1](cl. 5 (clause number to be checked against the owner's copy)); Hygrothermal performance of building components and elements — interstitial condensation (Glaser method)[1](cl. 6 (clause number to be checked against the owner's copy)); Internal humidity from the humidity class, moisture excess and its allowance (clause to be checked)[1]; Assessment of moisture transfer by numerical simulation (the whole standard, for timber and breathable builds)[2].

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
layers	Wall layers from the inside outwards (thickness d, conductivity λ and vapour resistance factor μ)	4 row(s)		drawing: A-310 rev C	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
climate	Twelve monthly outside conditions (the user's climate record, in the order the year is run)	12 row(s)		assumption: Invented illustrative maritime year; replace with the project's own monthly climate record (take it from your licensed source).	any
$humidity_{mode}$	Internal humidity from	constant internal relative humidity		assumption: Constant internal humidity; no humidity-class record is held for this project.	any
breathable	The build-up is timber or breathable (hygroscopic layers, capillary transport)	no		assumption: Mineral wool and brick: no timber or hygroscopic layers.	any
T_{int}	Internal air temperature	20	°C	spec: SPEC-A-02 §3.1: living-room design temperature	10 … 30 °C
RH_{int}	Internal relative humidity (constant-humidity mode)	50	%	spec: SPEC-A-02 §3.1: internal design humidity	20 … 90 %
Rsi	Internal surface resistance R_{si}	0.13	m²·K/W	spec: Project surface-resistance record (internal, horizontal flow), assumed	0.04 … 0.25 m²·K/W
Rse	External surface resistance R_{se}	0.04	m²·K/W	spec: Project surface-resistance record (external), assumed	0.02 … 0.25 m²·K/W
Rsi_f	Internal surface resistance $R_{si,f}$ for the f_{Rsi} check (0.25 m²K/W is the conventional value for an opaque element and 0.13 for glazing; a record to check against the owner's copy of ISO 13788)	0.25	m²·K/W	spec: Conventional R_si,f = 0.25 m²K/W for the surface-condensation check of an opaque element (a record, to be checked against the owner's copy of ISO 13788), assumed	0.04 … 0.5 m²·K/W

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
dp_f	Safety factor on the vapour-pressure difference $\Delta p = \Delta v \cdot R_v \cdot (T_i + T_e)/2$ in the moisture-excess mode (1.10 by convention; a record, wording to be checked against the owner's copy of ISO 13788)	1.1		default	1 … 1.5
RH_{crit}	Critical surface relative humidity	80	%	spec: SPEC-A-02 §4.4: critical surface humidity	50 … 100 %
Mc_{limit}	Limit on the condensate on any interface (optional; blank = report only)	500	g/m²	spec: SPEC-A-02 §4.4: limit on the condensate per interface	1 … 5000 g/m²

TABLE ROW	VALUES
layers[1]	name = Plasterboard; d = 12.5 mm; lambda = 0.25 W/(m·K); mu = 8
layers[2]	name = Vapour control layer; d = 0.3 mm; lambda = 0.2 W/(m·K); mu = 20000
layers[3]	name = Mineral wool; d = 50 mm; lambda = 0.035 W/(m·K); mu = 1
layers[4]	name = Existing brick; d = 215 mm; lambda = 0.8 W/(m·K); mu = 10
climate[1]	month = M1; theta_e = 2 °C; RH_e = 90 %
climate[2]	month = M2; theta_e = 3 °C; RH_e = 88 %
climate[3]	month = M3; theta_e = 6 °C; RH_e = 84 %
climate[4]	month = M4; theta_e = 9 °C; RH_e = 80 %
climate[5]	month = M5; theta_e = 13 °C; RH_e = 76 %
climate[6]	month = M6; theta_e = 16 °C; RH_e = 74 %
climate[7]	month = M7; theta_e = 18 °C; RH_e = 74 %
climate[8]	month = M8; theta_e = 18 °C; RH_e = 76 %
climate[9]	month = M9; theta_e = 15 °C; RH_e = 82 %
climate[10]	month = M10; theta_e = 11 °C; RH_e = 87 %
climate[11]	month = M11; theta_e = 6 °C; RH_e = 90 %
climate[12]	month = M12; theta_e = 3 °C; RH_e = 91 %

3. Assumptions

A1. Vapour diffuses one-dimensionally through homogeneous layers in series, in steady state each month. (The Glaser method.)
A2. Each month uses its mean climate, with no air movement, liquid transport, moisture storage, or solar or rain gain. (The Glaser method.)
A3. Material properties λ and μ are constants of each layer. (The Glaser method.)
A4. Condensate does not change the properties of a layer. (The Glaser method.)
A5. Condensation is checked at the interfaces between layers. (The Glaser method.)

A6. The saturation line is straight between interface values, which can miss condensate inside a layer. (Stated limit; whether ISO 13788 requires sub-division is to be checked against the owner's copy.)
A7. The annual balance is also run with every layer split into 20 sub-layers and passes only when both runs pass. (Refinement check.)
A8. The limit on the condensate per interface is checked on the interfaces as entered. (Stated limit.)
A9. The Magnus saturation pressure is used over water at and above 0 °C and over ice below. (Derived.)
A10. The water-vapour permeability of air is $\delta 0 = 2\times10^{-10}$ kg/(m·s·Pa). (Derived; closed-form physics, not data of a standard.)
A11. A blank month length is the calendar days of the month, with the climate rows taken as January to December. (Calendar months, February 28 days; enter the days where the record starts in another month.)
A12. A wet interface that dries within a month is dry from the next month. (Monthly steps.)
A13. The rates of a month are those of the wet state at its start. (Monthly steps.)
A14. The climate, humidity-class points, surface resistances, critical humidity and condensate limit are the user's own records. (No table from a standard is shipped.)
A15. climate (Twelve monthly outside conditions (the user's climate record, in the order the year is run)) is taken as 12 row(s). (Invented illustrative maritime year; replace with the project's own monthly climate record (take it from your licensed source).)
A16. $\text{humidity}_{\text{mode}}$ (Internal humidity from) is taken as constant internal relative humidity. (Constant internal humidity; no humidity-class record is held for this project.)
A17. breathable (The build-up is timber or breathable (hygroscopic layers, capillary transport)) is taken as no. (Mineral wool and brick: no timber or hygroscopic layers.)
A18. Month length: 12 row(s) of the climate gave no days, so the calendar days of the month are used with the rows taken as January to December (February 28 days). If the record starts in another month, enter the days of each row. (Calendar months.)

4. Working

[8] $R_1 = d / \lambda$ [1](cl. 6)
= Plasterboard: 12.5 mm / 0.25 W/(m·K)
= 0.05 m²·K/W
Plasterboard: thermal resistance

[9] $sd_1 = \mu \cdot d$ [1](cl. 6)
= Plasterboard: 8 × 12.5 mm
= 0.1 m
Plasterboard: diffusion-equivalent air layer thickness

[10] $R_2 = d / \lambda$ [1](cl. 6)
= Vapour control layer: 0.3 mm / 0.2 W/(m·K)
= 0.0015 m²·K/W
Vapour control layer: thermal resistance

[11] $sd_2 = \mu \cdot d$ [1](cl. 6)
= Vapour control layer: 20000 × 0.3 mm
= 6 m
Vapour control layer: diffusion-equivalent air layer thickness

[12] $R_3 = d / \lambda$ [1](cl. 6)
= Mineral wool: 50 mm / 0.035 W/(m·K)
= 1.4286 m²·K/W
Mineral wool: thermal resistance

[13] $sd_3 = \mu \cdot d$ [1](cl. 6)
= Mineral wool: 1 × 50 mm
= 0.05 m
Mineral wool: diffusion-equivalent air layer thickness

[14] $R_4 = d / \lambda$ [1](cl. 6)
= Existing brick: 215 mm / 0.8 W/(m·K)
= 0.26875 m²·K/W

Existing brick: thermal resistance

[15] $sd_4 = \mu \cdot d$ [1](cl. 6)

= Existing brick: 10 × 215 mm

= 2.15 m

Existing brick: diffusion-equivalent air layer thickness

[16] $R_T = R_{si} + \Sigma R_j + R_{se}$ [1](cl. 6)

= 0.13 + 1.7488 + 0.04

= 1.9188 m²·K/W

Total thermal resistance

[17] $U_{wall} = 1/R_T$ [1](cl. 6)

= 1 / 1.9188

= 0.52115 W/(m²·K)

U-value of the wall from the layers and the surface resistances above

[18] $S_d = \Sigma s_{d,j}$ [1](cl. 6)

= 8.3 m

= 8.3 m

Total diffusion-equivalent air layer thickness

[19] $psat_i = 610.5 \cdot \exp(17.269 \cdot \theta/(237.3 + \theta))$ [1](cl. 6)

= θ = 20 °C

= 2337 Pa

Saturation vapour pressure at the internal air temperature (Magnus)

[20] $p_{ext,m} = RH_e \cdot p_{sat}(\theta_e)$ [1](cl. 6)

= months 1 … 12 of the climate

= [634.76, 666.5, 785.09, 917.96, 1137.7, 1344.8, 1526.5, 1567.8, 1397.6, 1141.4, 841.16, 689

Outside vapour pressure by month

[21] $p_{int,m} = RH_i \cdot p_{sat}(\theta_i)$ [1](cl. 6)

= 50 % × 2337 Pa

= [1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 11

Inside vapour pressure by month

[22] n_{start} = first month of condensation [1](cl. 6)

= no month condenses

= 0

Row of the climate in which the accumulation starts (0 when nothing condenses)

[23] $Mc_1 = \Sigma$ Mc over the interfaces [1](cl. 6)

= M1: θ_e = 2 °C; 0 g/m² gained (-: dried) this month

= 0 g/m²

M1: condensate on the wall at the end of the month

[24] $Mc_2 = \Sigma$ Mc over the interfaces [1](cl. 6)

= M2: θ_e = 3 °C; 0 g/m² gained (-: dried) this month

= 0 g/m²

M2: condensate on the wall at the end of the month

[25] $Mc_3 = \Sigma$ Mc over the interfaces [1](cl. 6)

= M3: θ_e = 6 °C; 0 g/m² gained (-: dried) this month

= 0 g/m²

M3: condensate on the wall at the end of the month

[26] $Mc_4 = \Sigma$ Mc over the interfaces [1](cl. 6)

= M4: θ_e = 9 °C; 0 g/m² gained (-: dried) this month

= 0 g/m²

M4: condensate on the wall at the end of the month

[27] $Mc_5 = \Sigma$ Mc over the interfaces [1](cl. 6)

= M5: θ_e = 13 °C; 0 g/m² gained (-: dried) this month

= 0 g/m²

M5: condensate on the wall at the end of the month

[28] Mc_6 = Σ Mc over the interfaces [1](cl. 6)
= M6: θ_e = 16 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M6: condensate on the wall at the end of the month
[29] Mc_7 = Σ Mc over the interfaces [1](cl. 6)
= M7: θ_e = 18 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M7: condensate on the wall at the end of the month
[30] Mc_8 = Σ Mc over the interfaces [1](cl. 6)
= M8: θ_e = 18 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M8: condensate on the wall at the end of the month
[31] Mc_9 = Σ Mc over the interfaces [1](cl. 6)
= M9: θ_e = 15 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M9: condensate on the wall at the end of the month
[32] Mc_{10} = Σ Mc over the interfaces [1](cl. 6)
= M10: θ_e = 11 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M10: condensate on the wall at the end of the month
[33] Mc_{11} = Σ Mc over the interfaces [1](cl. 6)
= M11: θ_e = 6 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M11: condensate on the wall at the end of the month
[34] Mc_{12} = Σ Mc over the interfaces [1](cl. 6)
= M12: θ_e = 3 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M12: condensate on the wall at the end of the month
[35] $Mc_{if,1}$ = max over the year of Mc [1](cl. 6)
= between 'Plasterboard' and 'Vapour control layer'
= 0 g/m²
Most condensate held on the interface between 'Plasterboard' and 'Vapour control layer'
[36] $Mc_{if,2}$ = max over the year of Mc [1](cl. 6)
= between 'Vapour control layer' and 'Mineral wool'
= 0 g/m²
Most condensate held on the interface between 'Vapour control layer' and 'Mineral wool'
[37] $Mc_{if,3}$ = max over the year of Mc [1](cl. 6)
= between 'Mineral wool' and 'Existing brick'
= 0 g/m²
Most condensate held on the interface between 'Mineral wool' and 'Existing brick'
[38] Mc_{max} = max over interfaces and months of Mc [1](cl. 6)
= 0 g/m²
= 0 g/m²
Most condensate held on any one interface in the year
[39] Mc_{tot} = max over months of Σ Mc [1](cl. 6)
= 0 g/m²
= 0 g/m²
Most condensate on the whole wall in the year
[40] Mc_{end} = Σ Mc after 12 months [1](cl. 6)
= 0 g/m²
= 0 g/m²
Condensate left on the wall at the end of the 12-month cycle

[41] n_{diag} = month drawn
= M1 (row 1)
= 1
Row of the climate the diagram vectors are for
[42] sd_{if} = Σ s_d to the node [1](cl. 6)
= 5 nodes, inside surface first
= [0, 0.1, 6.1, 6.15, 8.3] m
Diffusion thickness at the surfaces and interfaces
[43] x_{if} = Σ d to the node [1](cl. 6)
= 5 nodes
= [0, 0.0125, 0.0128, 0.0628, 0.2778] m
Depth from the inside surface at the surfaces and interfaces
[44] $T_{if} = \theta_i - (\theta_i - \theta_e)\cdot(R_{si} + \Sigma R)/R_T$ [1](cl. 6)
= θ_e = 2 °C
= [18.781, 18.311, 18.297, 4.8963, 2.3752] degC
Temperature at the surfaces and interfaces
[45] $psat_{if}$ = 610.5·exp(17.269·θ/(237.3 + θ)), ice below 0 °C [1](cl. 6)
= 5 nodes
= [2166.3, 2103.6, 2101.7, 865.57, 724.45] Pa
Saturation vapour pressure at the surfaces and interfaces (the Glaser saturation line)
[46] p_{if} = tangent construction [1](cl. 6)
= 5 nodes
= [1168.5, 1162, 776.23, 773.01, 634.76] Pa
Vapour pressure at the surfaces and interfaces after the tangent construction
[47] $R_{Tf} = R_{si,f} + \Sigma R_j + R_{se}$ [1](cl. 5)
= 0.25 + 1.7488 + 0.04
= 2.0388 m²·K/W
Total thermal resistance with the internal surface resistance of the surface check
[48] $fRsi = 1 - R_{si}/R_T$ [1](cl. 5)
= 1 - 0.25 / 2.0388
= 0.87738
Temperature factor of the internal surface
[49] $fRsi_{min}$ = max over months of $(\theta_{si,min} - \theta_e)/(\theta_i - \theta_e)$ [1](cl. 5)
= $\theta_{si,min}$ from p_i/φ_{crit}; governing month: M1
= 0.59026
Minimum temperature factor at the critical surface humidity
[50] n_{fcrit} = month that governs $f_{Rsi,min}$ [1](cl. 5)
= M1 (row 1)
= 1
Row of the climate that governs the surface check (0 when none applies)
[51] $Mc_{end,fine}$ = Σ Mc after 12 months, each layer in 20 sub-layers [1](cl. 6)
= 0 g/m²
= 0 g/m²
Condensate left after 12 months when every layer is split into 20 sub-layers (refinement check)
[52] $Mc_{tot,fine}$ = max over months of Σ Mc, each layer in 20 sub-layers [1](cl. 6)
= 0 g/m²
= 0 g/m²
Most condensate on the whole wall when every layer is split into 20 sub-layers (refinement check)
[54] $u_{Mc} = Mc_{max} / Mc_{limit}$
= 0 g/m² / 500 g/m²
= 0
Utilisation of the limit on the condensate per interface

[55] u_f = $f_{Rsi,min}$ / f_{Rsi} [1](cl. 5)
= 0.59026 / 0.87738
= 0.67275
Utilisation of the surface temperature factor
[56] pass = every check passes
= 3 check(s)
= 1
1 when the annual balance, the limit and the surface factor all pass, else 0

5. Results

SYMBOL	VALUE	DESCRIPTION
Mc_{max}	0 g/m²	Most condensate on any interface
fRsi	0.87738	Temperature factor of the internal surface
pass	1	All checks pass
U_{wall}	0.5212 W/(m²·K)	U-value of the wall
R_T	1.919 m²·K/W	Total thermal resistance
S_d	8.300 m	Total diffusion-equivalent air layer thickness
$psat_i$	2337 Pa	Saturation vapour pressure at the internal temperature
$p_{ext,m}$	[634.76, 666.5, 785.09, 917.96, 1137.7, 1344.8, 1526.5, 1567.8, 1397.6, 1141.4, 841.16, 689.22] Pa	Outside vapour pressure by month
$p_{int,m}$	[1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5] Pa	Inside vapour pressure by month
n_{start}	0	First month of condensation
Mc_1	0 g/m²	M1: condensate on the wall
Mc_2	0 g/m²	M2: condensate on the wall
Mc_3	0 g/m²	M3: condensate on the wall
Mc_4	0 g/m²	M4: condensate on the wall
Mc_5	0 g/m²	M5: condensate on the wall
Mc_6	0 g/m²	M6: condensate on the wall
Mc_7	0 g/m²	M7: condensate on the wall
Mc_8	0 g/m²	M8: condensate on the wall
Mc_9	0 g/m²	M9: condensate on the wall
Mc_{10}	0 g/m²	M10: condensate on the wall
Mc_{11}	0 g/m²	M11: condensate on the wall
Mc_{12}	0 g/m²	M12: condensate on the wall
$Mc_{if,1}$	0 g/m²	Peak condensate on interface 1
$Mc_{if,2}$	0 g/m²	Peak condensate on interface 2

SYMBOL	VALUE	DESCRIPTION
$Mc_{if,3}$	0 g/m²	Peak condensate on interface 3
Mc_{tot}	0 g/m²	Most condensate on the wall
Mc_{end}	0 g/m²	Condensate left after 12 months
$Mc_{end,fine}$	0 g/m²	Condensate left after 12 months, each layer in 20 sub-layers
$Mc_{tot,fine}$	0 g/m²	Most condensate on the wall, each layer in 20 sub-layers
n_{diag}	1	Month of the diagram
sd_{if}	[0, 0.1, 6.1, 6.15, 8.3] m	Diffusion thickness at the nodes
x_{if}	[0, 0.0125, 0.0128, 0.0628, 0.2778] m	Depth at the nodes
T_{if}	[18.781, 18.311, 18.297, 4.8963, 2.3752] degC	Temperature at the nodes
$psat_{if}$	[2166.3, 2103.6, 2101.7, 865.57, 724.45] Pa	Saturation pressure at the nodes
p_{if}	[1168.5, 1162, 776.23, 773.01, 634.76] Pa	Vapour pressure at the nodes
$fRsi_{min}$	0.59026	Minimum temperature factor (governing)
n_{fcrit}	1	Month governing the surface check
R_{Tf}	2.039 m²·K/W	Total resistance for the surface check
u_{Mc}	0.00	Peak condensate against its limit
u_f	0.67	Surface factor against its minimum

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Minimum surface temperature factor against the factor of the wall at 80 % critical humidity	0.59026	0.87738	0.67	[1](cl. 5)
PASS	Condensate left after 12 months against zero	0 g/m² as entered, **0** g/m² (the wall dries out within the year)	—	—	[1](cl. 6)
PASS	Peak condensate on an interface against its limit	0 g/m²	500 g/m²	0.00	[1](cl. 6)

7. Applicability

WARNING records: tabulated values not taken from a reference record are layers (Layer conductivity and vapour resistance); Rsi, Rse, Rsi_f (Surface resistances); climate (Monthly outside climate); RH_{crit}, Mc_{limit} (Condensation criteria). Enter each as one so its source, edition and licence travel with it. [record]
WARNING layers[2].d: 0.3 mm is outside the usual range 1 … 500 mm (Thickness). [input.layers[2].d]

Defaults used (inputs left empty):
dp_f: not entered, so the default of 1.1 is used.

8. Verification

```
Template bs.glaser@1.4.0 Content hash a1230d60eae9a8db Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. BS EN ISO 13788:2012.
2. BS EN 15026:2007.